

HW#2

Thursday, September 10, 2026 7:10 PM

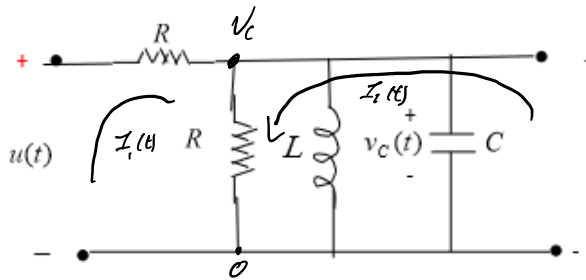
EEEE- 414: Classical Control

Homework Assignment # 2

Due date: by 11:59 PM, Thursday, September 10, 2026

Problem #1

For the following RLC circuit. Assume $R = 10K\Omega$, $L = 200H$, and $C = 10\mu F$



Find the capacitor voltage for the unit step input. Plot the capacitor voltage (use MATLAB)

$$\frac{u(s) - v_c(s)}{R} = \frac{v_c(s)}{R} + \frac{v_c(s)}{sL} + sC v_c(s)$$

$$u(s) - v_c(s) = v_c(s) + \frac{R v_c(s)}{sL} + sRC v_c(s)$$

$$u(s) - v_c(s) = v_c(s) \left(1 + \frac{R}{sL} + sRC \right)$$

$$v_c(s) + v_c(s) \left(\frac{R}{sL} + sRC \right) = u(s)$$

$$v_c(s) \left(1 + \frac{R}{sL} + sRC \right) = u(s)$$

$$v_c(s) = \frac{u(s)}{1 + \frac{R}{sL} + sRC}$$

$$\frac{v_c(s)}{u(s)} = \frac{1}{1 + \frac{R}{sL} + sRC}$$

$$\frac{v_c(s)}{u(s)} = \frac{sRC}{s^2 RC + \frac{R}{L} + s}$$

$$\frac{v_c(s)}{u(s)} = \frac{10s}{s^2 + 20s + 500}$$

$$v_c(s) = \frac{1}{s} \cdot \frac{10s}{s^2 + 20s + 500}$$

$$v_c(s) = \frac{10}{s^2 + 20s + 500}$$

```
% Q1
syms s t
V = 10 / (s^2 + 20 * s + 500);
ilaplace(V, s, t)
t = 0:0.001:0.5;
Vc = (sin(20 * t) .* exp(-10*t))/2;
figure
plot(t, Vc);
grid on
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$$V_c(s) = \frac{10}{s^2 + 20s + 500}$$

$$V_c(s) = \frac{10}{s^2 + 20s + 100 + 50}$$

$$V_c(s) = \frac{10}{(s+10)^2 + 20^2}$$

$$V_c(t) = \frac{1}{2} e^{-10t} \sin(20t)$$

$$vc = (\sin(20 \cdot t) \cdot \exp(-10 \cdot t)) / 2;$$

figure

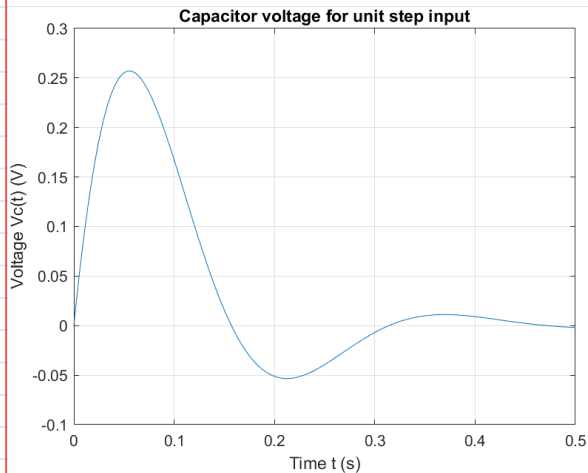
plot(t, Vc);

grid on

xlabel('Time t (s)')

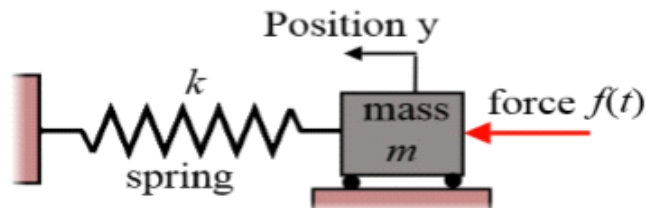
ylabel('Voltage Vc(t) (V)')

title('Capacitor voltage for unit step input')



Problem # 2

Consider the mass-spring system shown below. Find $y(t)$ when the input is $f(t) = u(t)$



$$m = 2 \text{ Kg}$$

$$k = 6 \text{ N/m}$$

$$v = \text{friction coefficient} = 2 \text{ Ns/m}$$

$$F_{\text{net}} = ma$$

$$F_{\text{net}} = m \frac{dy}{dt}$$

$$f(t) - f_s - f_d = m \frac{dy}{dt}$$

$$u(t) - k y(t) - c \frac{dy}{dt} = m \frac{d^2 y}{dt^2}$$

$$u(s) - k y(s) - c s y(s) + c y(0) = m (s^2 y(s) - s y(0) - y'(0))$$

$$u(s) - k y(s) - c s y(s) + 0 = m s^2 y(s) - 0 - 0$$

$$u(s) - y(s) (k + c s) = m s^2 y(s)$$

$$u(s) = m s^2 y(s) + y(s) (k + c s)$$

$$u(s) = y(s) (m s^2 + c s + k)$$

$$U(s) = ms^2 y(s) + y(s)(k + ms)$$

$$U(s) = y(s)(ms^2 + ms + k)$$

$$\frac{y(s)}{U(s)} = \frac{1}{ms^2 + ms + k}$$

$$y(s) = \frac{1}{s(ms^2 + ms + k)}$$

$$y(s) = \frac{1}{s(2s^2 + 2s + 6)}$$

$$y(s) = \frac{1}{2s(s^2 + s + 3)}$$

$$y(s) = \frac{1}{6s} - \frac{1}{6} \frac{s+1}{s^2 + s + 3}$$

$$y(s) = \frac{1}{6s} - \frac{1}{6} \frac{s + \frac{1}{2} + \frac{5}{4}}{s^2 + s + \frac{11}{4}}$$

$$y(s) = \frac{1}{6s} - \frac{1}{6} \frac{s + \frac{1}{2}}{(s + \frac{1}{2})^2 + \frac{1}{4}} - \frac{1}{6} \frac{\frac{5}{4}}{\frac{1}{4}((s + \frac{1}{2})^2 + \frac{1}{4})}$$

$$y(t) = \frac{1}{6} u(t) - \frac{1}{6} e^{-0.5t} (\cos(\sqrt{11/4} t) - \frac{1}{2} 0.3015 e^{0.5t} \sin(\sqrt{11/4} t))$$

$$y(t) = \frac{1}{6} u(t) - \frac{1}{6} e^{-0.5t} (\cos(\sqrt{11/4} t) + 0.3015 \sin(\sqrt{11/4} t))$$

% Q2

F = 1 / (2*s * (s^2 + s + 3));

partfrac(F, s)

t = 0:0.001:12;

Y = 1/6 - (1/6)*exp(-0.5*t).*(cos(sqrt(11/4)*t) + 0.3015*sin(sqrt(11/4)*t));

figure

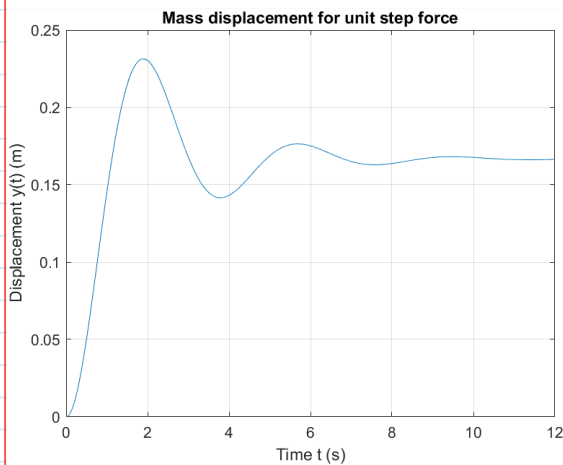
plot(t, Y);

grid on

xlabel('Time t (s)')

ylabel('Displacement y(t) (m)')

title('Mass displacement for unit step force')



Problem # 3

For each of the second order system shown below, find natural frequency, damping factor, rise time, settling time, time to peak, and OS.

a)

$$T(s) = \frac{16}{s^2 + 3s + 16}$$

b)

$$T(s) = \frac{0.04}{s^2 + 0.02s + 0.04}$$

c)

$$T(s) = \frac{1.05 \times 10^7}{s^2 + 1.6 \times 10^3 s + 1.05 \times 10^7}$$

A) $G(s) = \frac{2h^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$$\omega_n = \sqrt{16}$$

$$\omega_n = 4 \text{ rad/s}$$

$$2\zeta\omega_n = 3$$

$$2(\zeta) \cdot 4 = 3$$

$$\zeta = 3/8$$

$$T_s = 4/2\zeta\omega_n$$

$$T_s = 4/3/4 \cdot 4$$

$$T_s = 2.67 \text{ s}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-(3/8)^2}}$$

$$T_p = 0.8472 \text{ s}$$

$$OS = 100e^{-\pi \zeta / \sqrt{1-\zeta^2}}$$

$$OS = 100e^{-\pi \cdot 3/8 / \sqrt{1-(3/8)^2}}$$

$$OS = 28.06 \%$$

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\sqrt{1-\zeta^2} \omega_n}$$

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-(3/8)^2}}{3/8}\right)}{\sqrt{1-(3/8)^2} \cdot 4}$$

$$T_r = 0.527 \text{ s}$$

b)

$$\omega_n = \sqrt{0.04}$$

$$\omega_n = 0.2 \text{ rad/s}$$

$$0.02 = 2 \text{ Z} \omega_n$$

$$0.02 = 2 (0.2) Z$$

$$Z = 0.05$$

$$T_s = 4 / Z \omega_n$$

$$T_s = 4 / 0.05 (0.2)$$

$$T_s = 400 \text{ s}$$

$$T_p = \frac{\pi}{\sqrt{1-Z^2} \omega_n}$$

$$T_p = \frac{\pi}{\sqrt{1-0.05^2} (0.2)}$$

$$T_p = 15.728 \text{ s}$$

$$OS = 100 e^{-\pi Z / \sqrt{1-Z^2}}$$

$$OS = 100 e^{-\pi \cdot 0.05 / \sqrt{1-0.05^2}}$$

$$OS = 85.45 \%$$

$$T_r = \frac{\pi - \tan^{-1} \left(\frac{\sqrt{1-Z^2}}{Z} \right)}{\sqrt{1-Z^2} \omega_n}$$

$$T_r = \frac{\pi - \tan^{-1} \left(\frac{\sqrt{1-0.05^2}}{0.05} \right)}{\sqrt{1-0.05^2} \cdot 0.2}$$

$$T_r = 8.11 \text{ s}$$

c)

$$\omega_n = \sqrt{1.05} \times 10^4$$

$$\omega_n = 3240.37 \text{ rad/s}$$

$$1.6 \times 10^3 = 2 \text{ Z} \omega_n$$

$$1.6 \times 10^3 = 2 (3240.37) Z$$

$$Z = 0.247$$

$$T_s = 4 / Z \omega_n$$

$$T_s = 4 / 0.247 (3240.37)$$

$$T_s = 0.005 \text{ s}$$

$$T_p = \frac{\pi}{\sqrt{1-Z^2} \omega_n}$$

$$T_p = \frac{\pi}{\sqrt{1-0.247^2} (3240.37)}$$

$$T_p = 0.001 \text{ s}$$

$$OS = 100 e^{-\pi Z / \sqrt{1-Z^2}}$$

$$OS = 100 e^{-\pi (0.247) / \sqrt{1-0.247^2}}$$

$$OS = 100e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$$

$$OS = 100e^{-\frac{\pi(0.247)}{\sqrt{1-0.247^2}}}$$

$$OS = 44.90\%$$

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\sqrt{1-\zeta^2} \omega_n}$$

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-0.247^2}}{0.247}\right)}{\sqrt{1-0.247^2} (3240.87)}$$

$$T_r = 0.0006 \text{ s}$$

Problem # 4

Derive an equation for the settling time of a second order system with damping factor of one (critically damped system).

$$G_{cl}(s) = \frac{2\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = 1$$

$$G_{cl}(s) = \frac{2\omega_n^2}{s^2 + 2\omega_n s + \omega_n^2}$$

$$G_{cl}(s) = \frac{2\omega_n^2}{(s + \omega_n)^2}$$

$$C(s) = R(s)G_{cl}(s) = \frac{2\omega_n^2}{s(s + \omega_n)^2}$$

$$C(s) = \frac{1}{s} + \frac{-1}{s + \omega_n} + \frac{-2\omega_n}{(s + \omega_n)^2}$$

$$S(t) = 1 - e^{-\omega_n t} - 2\omega_n t e^{-\omega_n t}$$

$$1 - e^{-\omega_n t} - 2\omega_n t e^{-\omega_n t} = 0.98$$

$$x = \omega_n t$$

$$1 - e^{-x} - x e^{-x} = 0.98$$

$$e^{-x} + x e^{-x} = 0.02$$

$$x = 5.8339 \approx 6$$

$$T_s = x/\omega_n$$

$$T_s = 6/\omega_n$$

Problem # 5

An autonomous system has an orientation system with transfer function

$$\frac{\theta(s)}{\theta_{ref}(s)} = \frac{53.176}{4.6s^2 + 31.281s + 53.176}$$

Make a plot of $\theta(t)$ in response to $\theta_{ref}(t) = 3u(t)$

$$\frac{\theta(s)}{\theta_{ref}(s)} = \frac{11.56}{s^2 + 6.8s + 11.56}$$

$$\omega_n = \sqrt{11.56}$$

$$\omega_n = 3.4 \text{ rad/s}$$

$$\zeta = 2(3.4)Z$$

$$Z = 1$$

$$\theta(s) = \frac{3}{s} \cdot \frac{11.56}{(s + 3.4)^2}$$

$$\theta(s) = \frac{3}{s} - \frac{256}{(s + 17)^2} - \frac{16}{(s + 17)}$$

$$\theta(s) = \frac{3}{s} - \frac{10.2}{(s + 3.4)^2} - \frac{3}{s + 3.4}$$

$$\theta(t) = 3u(t) - 10.2te^{-3.4t} - 3e^{-3.4t}$$

% Q5

T = 3/s * (11.56/(s + 3.4)^2);

partfrac(T, s)

t = 0:0.001:2;

Theta = 3 - exp(-3.4*t).*(10.2*t + 3);

figure

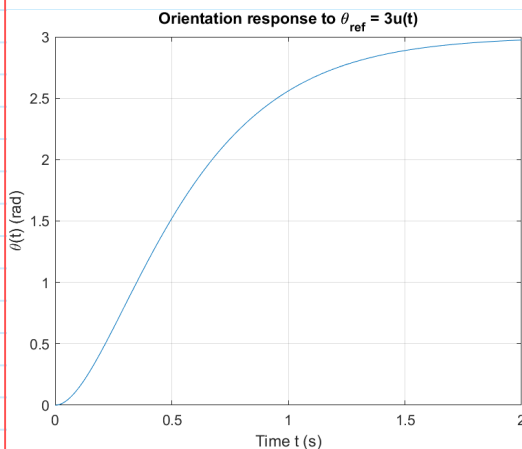
plot(t, Theta);

grid on

xlabel('Time t (s)')

ylabel('\theta(t) (rad)')

title('Orientation response to \theta_{ref} = 3u(t)')



Problem # 6

- Find the transfer function of a second-order system that yields a 15% overshoot and settling time of 0.7 second.
- Write an expression for the step response of the system.

$$A) \quad OS = 15\% \quad \tau_s = 0.7s$$

$$Z = \frac{-\ln(OS/100)}{\sqrt{\pi^2 + \ln(OS/100)^2}}$$

$$= \frac{211 + 1100i}{\sqrt{11^2 + 20^2}}$$

$$Z = \frac{-\ln(15/100)}{\sqrt{11^2 + 20^2}}$$

$$Z = 0.517$$

$$T_s = 1/2\omega_n$$

$$\omega_n = 1/Z$$

$$\omega_n = 1/0.517$$

$$\omega_n = 11.054 \text{ rad/s}$$

$$G_{cl}(s) = \frac{211}{s^2 + 22.24s + 211}$$

$$G_{cl}(s) = \frac{122.19}{s^2 + 2(0.517)(11.054)s + 122.19}$$

$$G_{cl}(s) = \frac{122.19}{s^2 + 11.93s + 122.19}$$

0)

$$C(s) = \frac{1}{s} \cdot \frac{122.19}{s^2 + 11.93s + 122.19}$$

$$C(s) = \frac{1}{s} - \frac{1008 + 1143}{1008s^2 + 1143s + 12219}$$

$$C(s) = \frac{1}{s} - \frac{s + 11.43}{s^2 + 11.43s + 122.19}$$

$$C(s) = \frac{1}{s} - \frac{s + 11.43}{(s^2 + 11.43s + 32.66) + 89.53}$$

$$C(s) = \frac{1}{s} - \frac{s + 5.715 + 5.715}{(s + 5.715)^2 + 89.53}$$

$$C(s) = \frac{1}{s} - \frac{s + 5.715}{(s + 5.715)^2 + 89.53} - \frac{5.715}{9.46} \cdot \frac{9.46}{(s + 5.715)^2 + 89.53}$$

$$C(t) = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - e^{-5.715t} \cos(9.46t) - 0.604 e^{-5.715t} \sin(9.46t)$$

% Q1

syms s t

V = 10 / (s^2 + 20 * s + 500);

ilaplace(V, s, t)

t = 0:0.001:0.5;

Vc = (sin(20 * t) .* exp(-10*t))/2;

figure

plot(t, Vc);

grid on

xlabel('Time t (s)')

ylabel('Voltage V_c(t) (V)')

title('Capacitor voltage for unit step input')

% Q2

F = 1 / (2*s * (s^2 + s + 3));

partfrac(F, s)

t = 0:0.001:12;

Y = 1/6 - (1/6)*exp(-0.5*t).*(cos(sqrt(11/4)*t) + 0.3015*sin(sqrt(11/4)*t));

```

figure
plot(t, Y);
grid on
xlabel('Time t (s)')
ylabel('Displacement y(t) (m)')
title('Mass displacement for unit step force')
% Q5
T = 3/s * (11.56/(s + 3.4)^2);
partfrac(T, s)
t = 0:0.001:2;
Theta = 3 - exp(-3.4*t).*(10.2*t + 3);
figure
plot(t, Theta);
grid on
xlabel('Time t (s)')
ylabel('\theta(t) (rad)')
title('Orientation response to \theta_{ref} = 3u(t)')
% Q6
C = 1/s * 122.19/(s^2 + 11.43*s + 122.19);
partfrac(C, s)

```